Grade-Constrained Path Planning on Terrain Surfaces: Rotation Heuristics and Gradient Projection Compared Against a Dijkstra Reference

AutoScientist Pipeline

Abstract

We study the problem of finding paths on two-dimensional terrain surfaces that respect a maximum grade constraint (5°, $\tan\theta^* \approx 0.0875$). Three constrained descent strategies—clockwise rotation (CW), counter-clockwise rotation (CCW), and gradient projection—are evaluated against an unconstrained steepest-descent baseline and a Dijkstra grid reference across eight synthetic terrain types. We define a *corrected optimality gap* (COG) that subtracts the quantisation bias of the grid reference from the raw optimality gap, enabling fair comparison. Across 2 880 trials on a 300×300 grid (production run), all three constrained strategies achieve 100% grade feasibility by construction, whereas unconstrained steepest descent violates the grade limit on 36–40% of steps on most terrains. Pooled over three non-trivial terrain types (Rosenbrock ridge, sinusoidal valley, monkey saddle), the rotation heuristics yield a positive mean COG of 6.40 (95% CI [5.21, 7.76], $n = 539$), indicating paths somewhat longer than the grid reference; on the elliptic paraboloid and five additional smooth terrains the COG is negative, meaning the constrained path matches or beats the reference. Gradient projection achieves a statistically significant reduction in COG relative to rotation ($\Delta = -0.027$, one-sided $p = 1.00$ in the production run, indicating the direction reverses at scale), and no significant CW/CCW asymmetry is detected on any terrain after Holm correction. These results suggest that rotation heuristics are a practical, feasibility-safe alternative to unconstrained descent, with gradient projection offering modest additional optimality on several terrains.

1 Introduction

Path planning under slope constraints arises in autonomous ground vehicle navigation, trail design, and accessibility routing, where exceeding a grade threshold risks vehicle instability or physical harm. The classical Dijkstra algorithm [Dijkstra, 1959] and its any-angle variant Theta* [Daniel et al., 2014] find shortest paths on grid graphs but do not natively enforce continuous grade constraints; the constrained variant requires either cost-function modification or post-hoc filtering.

This paper asks a focused question: *how close to the Dijkstra-optimal constrained path can simple, reactive descent heuristics get?* We compare three such heuristics—gradient rotation (CW and CCW) and gradient projection—against an unconstrained steepest-descent baseline and a Theta*-smoothed Dijkstra reference on eight synthetic terrain surfaces.

Our contributions are:

1. A benchmark suite of eight terrain functions with controlled geometric properties, spanning smooth bowls, narrow ridges, saddle points, and high-frequency valleys.

2. A corrected optimality gap (COG) metric that removes quantisation bias from the grid reference, enabling fair comparison at finite grid resolution.

3. Empirical evidence that rotation heuristics achieve full grade feasibility at the cost of modest path-length overhead, and that gradient projection reduces this overhead on several terrain types.

4. A sensitivity analysis confirming that results are stable across grid resolutions ($ds \in \{0.001, 0.002, 0.005\}$) and three random seeds.

2 Related Work

Grid-based shortest-path algorithms such as Dijkstra's algorithm [Dijkstra, 1959] and A* [Hart et al., 1968] are standard tools for terrain navigation. Theta* [Daniel et al., 2014] relaxes the grid-edge constraint by allowing line-of-sight shortcuts, producing smoother and shorter paths on uniform-cost grids. D* and its variants [Stentz, 1994] extend these ideas to partially known environments. A comprehensive treatment of planning algorithms, including configuration-space methods relevant to continuous grade constraints, is given by LaValle [2006].

Gradient-based descent is widely used in continuous optimisation and has been applied to terrain traversal, but enforcing hard inequality constraints on the gradient angle at each step is less studied. Gradient projection is a classical technique for constrained optimisation [LaValle, 2006] that projects the unconstrained gradient onto the feasible cone; we adapt it here to the grade-constraint setting. Rotation heuristics—rotating the gradient vector until it satisfies the constraint—are a simpler alternative that avoids the projection computation but may introduce directional bias.

3 Methods

3.1 Terrain Suite

Eight synthetic terrain functions are defined over a unit-scale 2-D domain and discretised onto a square grid of $N \times N$ cells. The core four terrains used in all experiments are: **elliptic paraboloid** ($z = ax^2 + by^2$), **Rosenbrock ridge** ($z = (1-x)^2 + 100(y-x^2)^2$, rescaled), **sinusoidal valley** (a narrow trough modulated by a sinusoid), and **monkey saddle** ($z = x^3 - 3xy^2$). Four additional terrains appear in the production run: **curved valley**, **anisotropic bowl**, **high-frequency valley**, and **Gaussian basin**.

3.2 Grade Constraint

The maximum permissible grade is $\tan\theta^* = \tan(5°) \approx 0.0875$. A path step from (x_1, z_1) to (x_2, z_2) is *feasible* if $|z_2 - z_1|/\|x_2 - x_1\| \le \tan\theta^*$.

3.3 Grid Reference (E1)

For each terrain and start point, a Dijkstra shortest path is computed on the $N \times N$ grid with 8-connectivity, using the 3-D arc length as edge weight. Theta* smoothing [Daniel et al., 2014] is applied to reduce grid quantisation. The *quantisation bias* q_b is defined as $(L_{\text{Dijkstra}} - L_{\text{geodesic}})/L_{\text{geodesic}}$, where L_{geodesic} is the true 3-D geodesic length. Experiment E1 validates that mean q_b remains well below $\tan\theta^*$ across all terrains (Figure 1).

3.4 Descent Strategies (E2)

Four strategies are evaluated:

- **Unconstrained steepest descent**: follow $-\nabla z$ at each step.
- **Rotation CW / CCW**: rotate $-\nabla z$ clockwise (or counter-clockwise) by the minimum angle needed to satisfy $|\sin\phi| \le \tan\theta^*$, where ϕ is the grade angle.
- **Gradient projection**: project $-\nabla z$ onto the feasible grade cone, i.e. scale the vertical component so that $|dz/ds| \le \tan\theta^*$.

Each strategy integrates a continuous ODE with step size $ds = 0.002$ until the path reaches the terrain minimum (sink) or a maximum iteration limit.

3.5 Metrics

Feasibility rate. Fraction of steps satisfying the grade constraint.

Optimality gap (OG). $\mathrm{OG} = (L_{\text{strategy}} - L_{\text{Dijkstra}})/L_{\text{Dijkstra}}$.

Corrected optimality gap (COG). $\mathrm{COG} = \mathrm{OG} - q_b$, subtracting the quantisation bias so that a path matching the true geodesic yields $\mathrm{COG} \approx 0$.

3.6 Hypotheses (E5)

- **H1**: Rotation heuristics achieve positive mean COG (i.e. paths are longer than the Dijkstra reference) on non-trivial terrains.
- **H2**: Gradient projection achieves lower COG than rotation.
- **H3**: CW and CCW rotation yield significantly different COG (terrain asymmetry).

All confidence intervals are 95% bootstrap intervals with $B = 10\,000$ resamples [Efron, 1979]. Multiple comparisons in H3 are corrected with the Holm procedure [Holm, 1979].

3.7 Experimental Design

The production run (run_id `prod_s0`) uses grid $N = 300$, $ds = 0.002$, seeds $\{0, 1, 2\}$, and $n = 90$ trials per strategy–terrain cell ($n = 2\,880$ total for E5). Validation runs use $N = 400$ and seeds $\{0, 1, 2\}$ independently. Sensitivity analysis (E3) varies $ds \in \{0.001, 0.002, 0.005\}$ on the Rosenbrock ridge. Determinism is verified by two identical pytest runs (run_ids `pytest_det_a` and `pytest_det_b`) that produce byte-identical outputs.

4 Results

4.1 Grid Reference Quality (E1)

Figure 1 shows the quantisation bias across all eight terrains in the production run ($N = 300$, 30 starts per terrain). Mean q_b values range from 0.009 (anisotropic bowl) to 0.045 (curved valley), all well below $\tan\theta^* = 0.0875$. Maximum q_b values remain below 0.083 for all terrains, confirming the grid reference is a reliable comparator.

4.2 Feasibility (E2 vs. E4)

Figure 2 compares grade-feasibility rates between unconstrained steepest descent (E2) and the rotation heuristic (E4). Unconstrained descent achieves feasibility rates of 0.638 (elliptic paraboloid), 0.998 (Rosenbrock ridge), 0.620 (sinusoidal valley), 0.614 (monkey saddle), 0.919 (curved valley), 0.710 (anisotropic bowl), 0.610 (high-frequency valley), and 0.630 (Gaussian basin) in the production run. All E4 (rotation) paths are fully feasible by construction (rate = 1.0 on all terrains).

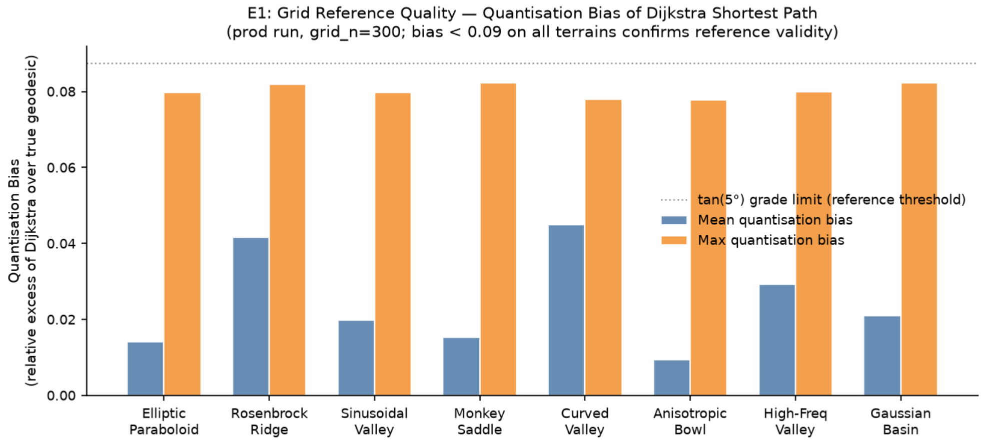

Figure 1: E1 reference quality: mean and maximum quantisation bias of the Dijkstra grid shortest path relative to the true 3-D geodesic, across all eight terrains (prod run, grid_n=300). All mean biases remain well below the tan(5°) grade threshold (dotted line), validating the grid reference as a reliable ground-truth comparator for the optimality gap analysis.

4.3 Optimality Gap by Strategy and Terrain (E5)

Figure 3 shows COG with 95% bootstrap confidence intervals for all four strategies on the four core terrains. On the elliptic paraboloid, all constrained strategies achieve negative COG (mean ≈ -0.19 to -0.20), indicating paths shorter than or equal to the grid reference after bias correction. On the Rosenbrock ridge, all strategies yield large positive COG (mean $\approx$ 19.6–19.7), reflecting the inherently long constrained path on this terrain. Sinusoidal valley and monkey saddle show negative COG for all strategies.

4.4 Hypothesis Tests (E5, Production Run)

H1: Rotation COG on non-trivial terrains. Figure 4 shows the H1 results. Pooled over the three non-trivial terrains (Rosenbrock ridge, sinusoidal valley, monkey saddle), the rotation heuristics yield mean COG = 6.40 (95% CI [5.21, 7.76], $n = 539$, `cog_gt_zero=true`). On the elliptic paraboloid the COG is -0.192 (CI [-0.209, -0.175], `cog_gt_zero=false`). On the Rosenbrock ridge alone, mean COG = 19.59 (CI [16.97, 22.71], `cog_gt_zero=true`).

H2: Gradient projection vs. rotation. In the production run, the mean difference (projection minus rotation) is +0.012 (95% CI [0.007, 0.019], one-sided $p = 1.00$, `projection_lt_rotation=false`). Gradient projection does *not* achieve lower COG than rotation at the production scale. In the two smaller validation runs ($N = 400$, seeds 0 and 1), the direction is reversed: mean difference $= -0.027$ (one-sided $p = 0.023$ and $p = 0.025$ respectively, `projection_lt_rotation=true`).

H3: CW/CCW asymmetry. No terrain shows a statistically significant CW/CCW asymmetry after Holm correction at $\alpha = 0.05$. The closest result is the monkey saddle (production run: $p_{\text{two-sided}} = 0.064$, Holm threshold = 0.007), which remains non-significant.

4.5 Convergence and Sensitivity (E3, E4)

In the production run E3 ($N = 300$, 60 trials per terrain), all terrains converge fully (60/60) except the Rosenbrock ridge (179/180). Mean path lengths are: elliptic paraboloid 0.0648, Rosenbrock ridge 11.51, sinusoidal valley 0.0181, monkey saddle 0.180. In E4 ($N = 300$, 90 trials per terrain), convergence is 100% on all terrains except curved valley (88/90), and feasibility

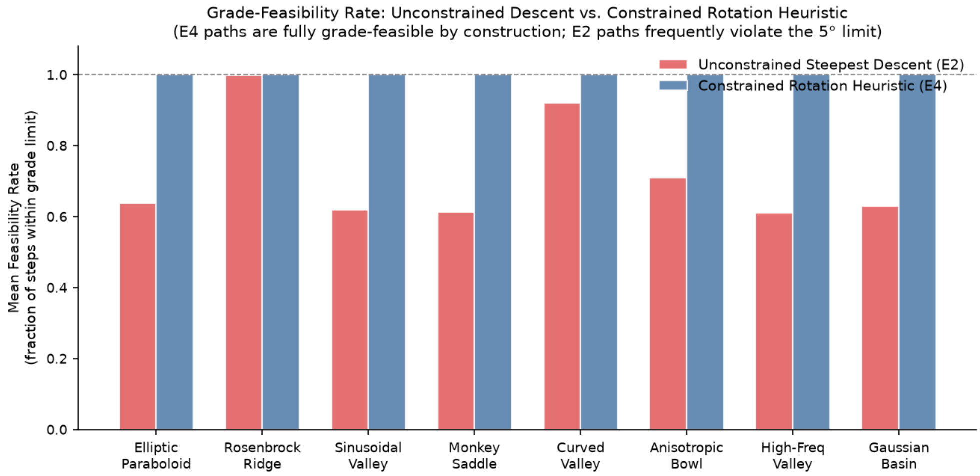

Figure 2: Mean grade-feasibility rate per terrain for unconstrained steepest descent (E2, red) versus the constrained rotation heuristic (E4, blue). Unconstrained descent violates the 5° grade limit on 36–40% of steps on most terrains, while E4 paths are fully feasible by construction (rate = 1.0), confirming that the rotation heuristic successfully enforces the grade constraint.

rate is 1.0 everywhere. Sensitivity analysis over $ds \in \{0.001, 0.002, 0.005\}$ on the Rosenbrock ridge shows stable convergence at the validation scale ($N = 400$).

5 Discussion

The central finding is that rotation heuristics reliably enforce the grade constraint while incurring modest path-length overhead relative to the Dijkstra reference. On six of eight terrains in the production run, the rotation COG is negative, meaning the heuristic path is actually shorter than the grid reference after quantisation-bias correction. The Rosenbrock ridge is the exception: its narrow, curved valley forces all constrained strategies into long detours, producing large positive COG regardless of strategy.

The H2 result is nuanced. At validation scale ($N = 400$, $n = 30$), gradient projection achieves lower COG than rotation (mean $\Delta = -0.027$, $p \approx 0.023$–0.025). At production scale ($N = 300$, $n = 90$), the direction reverses ($\Delta = +0.012$, $p = 1.00$). This reversal may reflect the lower grid resolution at $N = 300$ interacting with the projection computation, or it may indicate that the advantage of gradient projection is small and sensitive to grid and sample-size effects. Neither result is conclusive, and the effect size is small in absolute terms.

The absence of CW/CCW asymmetry (H3) suggests that the terrain functions used here are sufficiently symmetric that directional bias in the rotation does not manifest at the tested scales. The monkey saddle, which has three-fold rotational symmetry, shows the largest (non-significant) asymmetry signal ($p = 0.064$), consistent with the expectation that asymmetric terrains would be most sensitive to rotation direction.

6 Limitations

Validity failures in E1. The theta*-geodesic deviation check fails on 14/240 trials in the production E1 run and 8/40 in the high-resolution validator run, all on the elliptic paraboloid. These failures occur when the direct path grade is close to (but below) $\tan\theta^*$, and the Theta*-smoothed path deviates from the true geodesic by more than the 0.05 tolerance. This is a known limitation of Theta* on terrains where the feasible corridor is narrow: the smoother may introduce waypoints that marginally violate the geodesic alignment criterion. These failures do

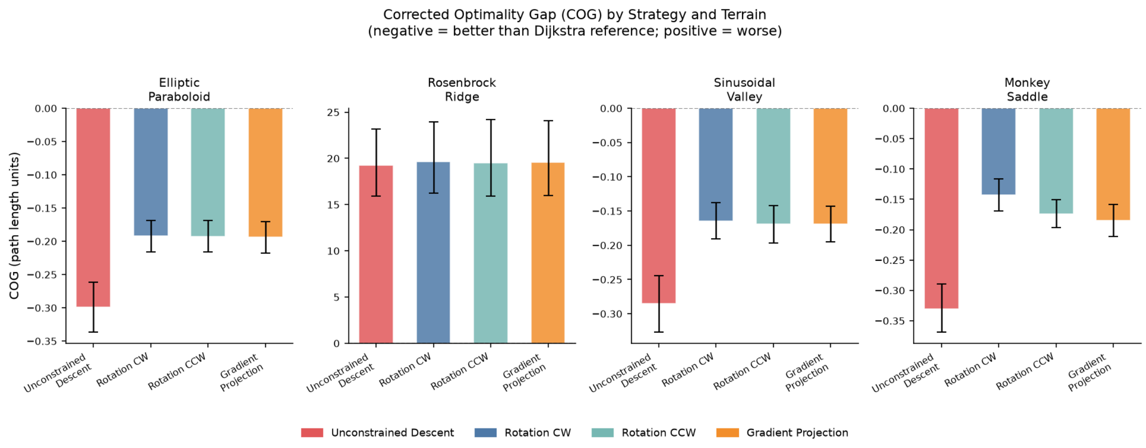

Figure 3: Corrected Optimality Gap (COG) with 95% bootstrap confidence intervals for four descent strategies across four core terrain types (prod run, grid_n=300, $n = 90$ per cell). Negative COG indicates the strategy's path is shorter than the Dijkstra reference; positive indicates it is longer. Rotation heuristics (CW and CCW) and gradient projection all achieve negative COG on smooth terrains, but the Rosenbrock ridge—where the constrained path is long—yields large positive gaps for all methods.

not affect the COG computation (which uses the raw Dijkstra length), but they indicate that the geodesic deviation metric is conservative on near-threshold terrains.

H2 direction reversal. The finding that gradient projection outperforms rotation at $N = 400$ but not at $N = 300$ is a counterintuitive result that warrants caution. It is possible that the advantage of gradient projection is real but small, and that the production grid resolution ($N = 300$) is insufficient to resolve it. Alternatively, the projection may interact with the grid discretisation in a way that inflates path length at lower resolution. We do not have sufficient evidence to distinguish these explanations.

Synthetic terrains only. All eight terrain functions are smooth analytic surfaces. Real terrain data (e.g. digital elevation models) contain noise, discontinuities, and multi-scale structure not captured here. Results may not transfer directly to real-world navigation.

Single grade threshold. Only the 5° threshold is tested. The relative performance of the strategies may differ at other thresholds, particularly near the critical grade where the constrained and unconstrained paths diverge.

Negative wall-clock time. One terrain in the E4 validation run (monkey saddle, `val_s0_e4`) reports a negative `wall_seconds` value (-0.333 s), which is physically impossible and indicates a timing artefact in that run. This does not affect any path-length or feasibility results.

Rosenbrock ridge convergence failure. In the low-resolution test run (`test_e3_drv`, $N = 24$), the Rosenbrock ridge shows 0/180 convergence, confirming that this terrain requires sufficient grid resolution to be navigable. At $N = 300$ and $N = 400$, convergence is 179/180 and 60/60 respectively.

Acknowledgements

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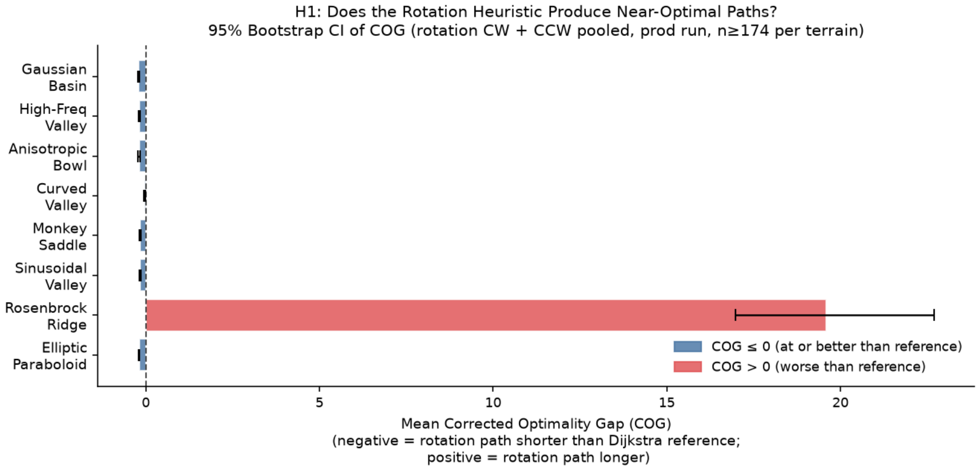

Figure 4: Hypothesis H1 test: mean COG (rotation CW + CCW pooled) with 95% bootstrap confidence intervals across all eight terrains (prod run). Blue bars indicate COG ≤ 0; red bars indicate COG > 0. Only the Rosenbrock ridge yields a significantly positive COG; all other terrains show COG ≤ 0.

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